

Carlos Daniel Carrasco

University of California, San Francisco & San Francisco VA Medical Center

carlos.carrasco@ucsf.edu

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EDUCATION

Ph.D., Neuroscience, University of California, Davis 2024
Advisors: Steven J. Luck, Ph.D. & Erie D. Boorman, Ph.D.

B.S., Neuroscience, University of Nevada, Reno 2017
Advisor: Marian Berryhill, Ph.D.

PROFESSIONAL AND ACADEMIC APPOINTMENTS

Postdoctoral Scholar, University of California, San Francisco & San Francisco VA 2025–Present
Brain Imaging & EEG Lab (Mathalon Lab)

Research Data Specialist, Office of Energy Infrastructure Safety 2024–2025

Graduate Student Researcher/Fellow, Teaching Assistant, UC Davis 2018–2024
Center for Mind & Brain (Luck Lab)

Jr. Research Specialist, University of Nevada, Reno 2018
Memory and Brain Lab (Berryhill Lab)

Student Research Assistant, University of Nevada, Reno 2015–2017
Memory and Brain Lab (Berryhill Lab); Neuroplasticity Lab (Jiang Lab)

PUBLICATIONS

- [1] Carrasco, C. D., Berryhill, M., & Jones, K. T. (2025). Older adults are impaired by distractors presented during working memory encoding. *Psychological Research*, 89, 160. <https://doi.org/10.1007/s00426-025-02197-y>
- [2] Gowda, H. T., Kaul, N., Carrasco, C. D., Batraw, M., Amer, S., Kotwal, S., Lam, S., et al. (2025). A database of upper limb surface electromyogram signals from demographically diverse individuals. *Scientific Data*, 12(1), 517. <https://doi.org/10.1038/s41597-025-04825-z>
- [3] Zhang, G., Carrasco, C. D., Winsler, K., Bahle, B., Cong, F., & Luck, S. J. (2024). Assessing the effectiveness of spatial PCA on SVM-based decoding of EEG data. *NeuroImage*, 293, 120625. <https://doi.org/10.1016/j.neuroimage.2024.120625>
- [4] Carrasco, C. D., Bahle, B., Simmons, A. M., & Luck, S. J. (2024). Using multivariate pattern analysis to increase effect sizes for event-related potential analyses. *Psychophysiology*, 61(7), e14570. <https://doi.org/10.1111/psyp.14570>
- [5] Carrasco, C. D., Simmons, A. M., Kiat, J. E., & Luck, S. J. (2025). Enhanced working memory representations for rare events. *Psychophysiology*, 62(3), e70038. <https://doi.org/10.1111/psyp.70038>
- [6] Shires, J., Carrasco, C. D., & Berryhill, M. (2020). No tDCS augmented working memory training benefit in undergraduates rewarded with course credit. *Brain Stimulation*, 13(6), 1524–1526. <https://doi.org/10.1016/j.brs.2020.08.015>

- [7] Vercillo, T., **Carrasco, C. D.**, & Jiang, F. (2017). Age-related changes in sensorimotor temporal binding. *Frontiers in Human Neuroscience*, 11, 500. <https://doi.org/10.3389/fnhum.2017.00500>

MANUSCRIPTS IN PREPARATION

Carrasco, C. D., Light, G. A., ... Mathalon, D. H. (in preparation). *Reduced P300 response with condition-specific habituation differences in clinical high-risk for psychosis: Convergent evidence from ERP amplitude and multivariate decoding.*

Carrasco, C. D., ... Mathalon, D. H. (in preparation). *Alpha-band desynchronization as a candidate EEG marker of psychosis risk in clinical high-risk youth.*

Wong, M. *, **Carrasco, C. D.** *, ... Mathalon, D. H. (in preparation). *A somatosensory oddball P300 marker of psychosis risk.*

*Co-first authors.

TEACHING EXPERIENCE

PSC 001Y, General Psychology — Teaching Assistant, UC Davis	2021, 2024
PSC 133, NeuroEconomics — Teaching Assistant, UC Davis	2021
NSC 298, Neuroimaging Journal Club — Discussion Leader, UC Davis	2018–2019
Neuroscience Boot Camp — Live EEG Demonstration (built demo software), UC Davis	2019–2023
SCI 110, First Year Experience — Discussion Leader, UNR	2015

MENTORSHIP

Neha Kaul (B.S. Cognitive Neuroscience, UC Davis, 2025)

Certified Clinical Medical Assistant (CCMA)

Saniya Kotwal (B.S. Cognitive Science, UC Davis, 2025)

Associate Product Manager, ShipStation Global

Safa Amer (B.S. Neurobiology, Physiology, and Behavior, UC Davis, 2025)

Behavior Specialist

Sebastian Lopez (B.S. Cognitive Neuroscience, UC Davis, 2024)

Ph.D. Student, University of Nevada, Reno

Samantha Fong (B.S. Cognitive Science, UC Davis, 2023)

Ph.D. Student, UC Santa Cruz

Emma Ratnaparkhi (B.S. Neurobiology, Physiology, and Behavior, UC Davis, 2022)

M.D. Student, Frank H. Netter MD School of Medicine, Quinnipiac University

Seigo Ninomiya (B.S. Cognitive Science, UC Davis, 2022)

Clinical Research Coordinator, Stanford University

Charlize Mitra (B.S. Neurobiology, Physiology, and Behavior, UC Davis, 2022)

Nursing Student, Samuel Merritt University

GRANT-SUPPORTED RESEARCH

Doctoral research (Luck Lab, UC Davis; PI: Steven J. Luck, Ph.D.)

R01EY033329 (National Eye Institute); R01MH087450 and R01MH076226 (National Institute of Mental Health)

Funded my published EEG/ERP decoding work in NeuroImage (2024) and Psychophysiology (2024, 2025).

Postdoctoral research (Mathalon Lab, UCSF & San Francisco VA; PI: Daniel H. Mathalon, Ph.D., M.D.)

Accelerating Medicines Partnership® Schizophrenia (AMP-SCZ) Program; U01MH076989 (National Institute of Mental Health; NAPLS)

Support my in-preparation manuscripts on EEG markers of psychosis risk.

RESEARCH GRANTS AND FELLOWSHIPS

NIH/NIMH T32 Training Grant in Clinical Neuroscience Trainee, University of California, Davis	2019–2020
NIH/NIMH T32 Training Grant in Cognitive Neuroscience Trainee, University of California, Davis	2018–2019
National Science Foundation (NSF) Undergraduate Research Opportunities Program (UROP) Awardee, University of Nevada, Reno	2017
Nevada Undergraduate Research Award Recipient, University of Nevada, Reno	2016
Ronald E. McNair Postbaccalaureate Achievement Program McNair Scholar, University of Nevada, Reno	2016–2017

HONORS AND AWARDS

Student Seminar Series Lead Speaker (twice selected) — UC Davis Cognitive Neuroscience Area	2022, 2023
DataBlitz Speaker — Cognitive Neuroscience Society, San Francisco, CA	2022
Graduate Student Association Travel Award — UC Davis	2021, 2023
Learning, Memory, and Plasticity (LaMP) NIH Fellow — UC Davis	2019–2020
Initiative for Maximizing Student Development (IMSD) NIH Fellow — UC Davis	2018–2019

SERVICE AND OUTREACH

Undergraduate Speaker, Advancing Diversity in Neuroscience Research — UC Davis,	2024
Undergraduate Mentor, Advancing Diversity in Neuroscience Research — UC Davis,	2023
McNair Scholars Program Speaker & Panelist (graduate school & research paths) — UC Davis,	2021
Neuroscience Graduate Group Peer Mentor — UC Davis,	2020
NeuroFest Organizer — UC Davis,	2020
Elementary School Brain Awareness Outreach — UC Davis (2018–2019); UNR (2017)	

INVITED TALKS

Dashboarding and data science for decision-making. California Natural Resources Agency, Sacramento, 2024

Multivariate EEG approaches, working memory, and contingency learning. Cognitive Brain Sciences Early Career Seminar Series, UNR, 2023

Are rare stimuli that produce a P3b component preferentially encoded in working memory?.

Cognitive Neuroscience Society, San Francisco, 2022

A comparison of ERPs, SVM decoding, and cross Mahalanobis distance analysis in EEG. UCD EEG Meeting, UC Davis, 2021

Behavioral oscillations in working memory. Ronald McNair Scholars Symposium, UC Berkeley, 2017

CONFERENCE POSTERS

Carrasco, C. D., Light, G. A., ... Mathalon, D. H. (2026). Neural habituation deficits in clinical high-risk youth [Conference abstract]. *Biological Psychiatry*, 99, S416–S417. <https://doi.org/10.1016/j.biopsych.2025.08.001>

Kaul, N.^{*}, Amer, S.^{*}, Kotwal, S.^{*}, Gowda, H. T., **Carrasco, C. D.**, & Miller, L. M. (2024). Towards feasible EEG-based BCIs for hand gesture control [Poster presentation]. UC Davis Undergraduate Research Conference, Davis, CA

Kaul, N.^{*}, Amer, S.^{*}, Kotwal, S.^{*}, Gowda, H. T., **Carrasco, C. D.**, & Miller, L. M. (2024). Towards feasible EEG-based BCIs for hand gesture control [Poster presentation]. Center for Mind and Brain Annual Poster Day, UC Davis, Davis, CA

^{*}Undergraduate mentee

Emergent model-based replay as a function of human credit assignment [Poster presentation]. Society for Neuroscience, Washington, DC, 2023

Rewards elicit choice reactivation for credit assignment [Poster presentation]. Society for Neuroscience, San Diego, 2022

Working memory and influences of selective attention [Poster presentation]. Nevada Undergraduate Research Symposium, UNR, 2017

Age-related changes in sensorimotor temporal binding [Poster presentation]. International Multisensory Research Forum, Vanderbilt University, 2017

Working memory and selective attention [Poster presentation]. Psi Chi Annual Research Symposium, UNR, 2016

SKILLS

Programming: Python, R, MATLAB, Bash

Neural data & toolboxes: EEGLAB/ERPLAB, MNE-Python, FieldTrip, BIDS, fMRIPrep

Methods: EEG/ERP, machine learning, multivariate decoding (LDA/SVM, MVPA), PCA/tPCA, mixed-effects modeling

Computing: Linux/Unix, high-performance and cloud computing (SLURM, SGE), Git/GitHub, VS Code

Software: PsychoPy, JASP, Microsoft Office